Experiment :- 1

Problem statement:-

Design and simulate a simple computer network using various connection topologies, namely bus, star, ring, and mesh. Compare the advantages and disadvantages of each topology in terms of data flow and network efficiency. Analyze the performance of these topologies under different network conditions.

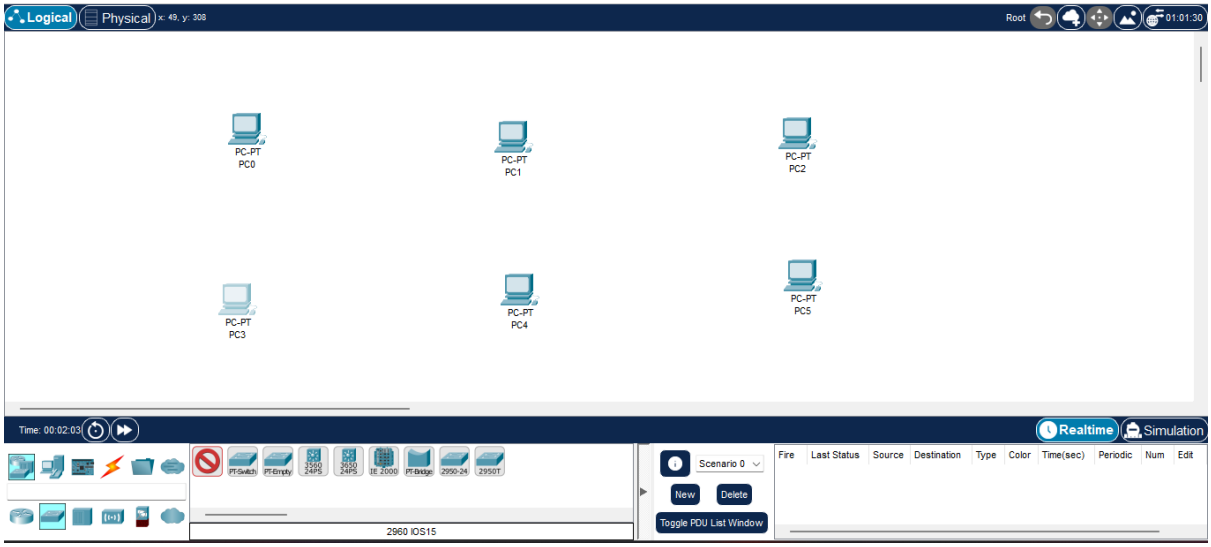
Aim:- Design and Simulation of Computer Network Topologies

Procedure:-

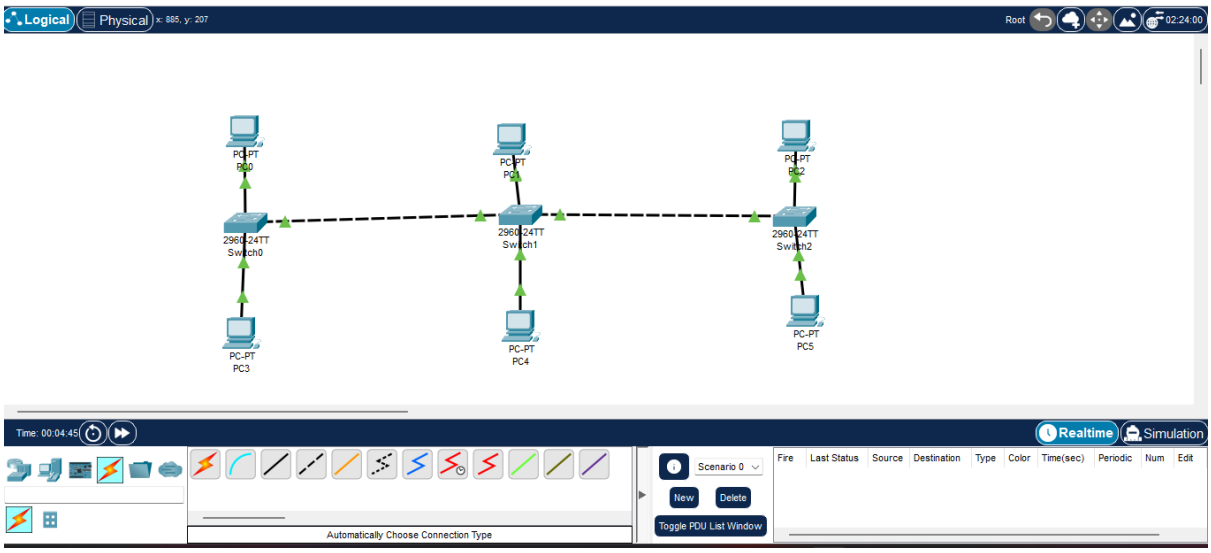
1. **Bus topology:-**

Steps:

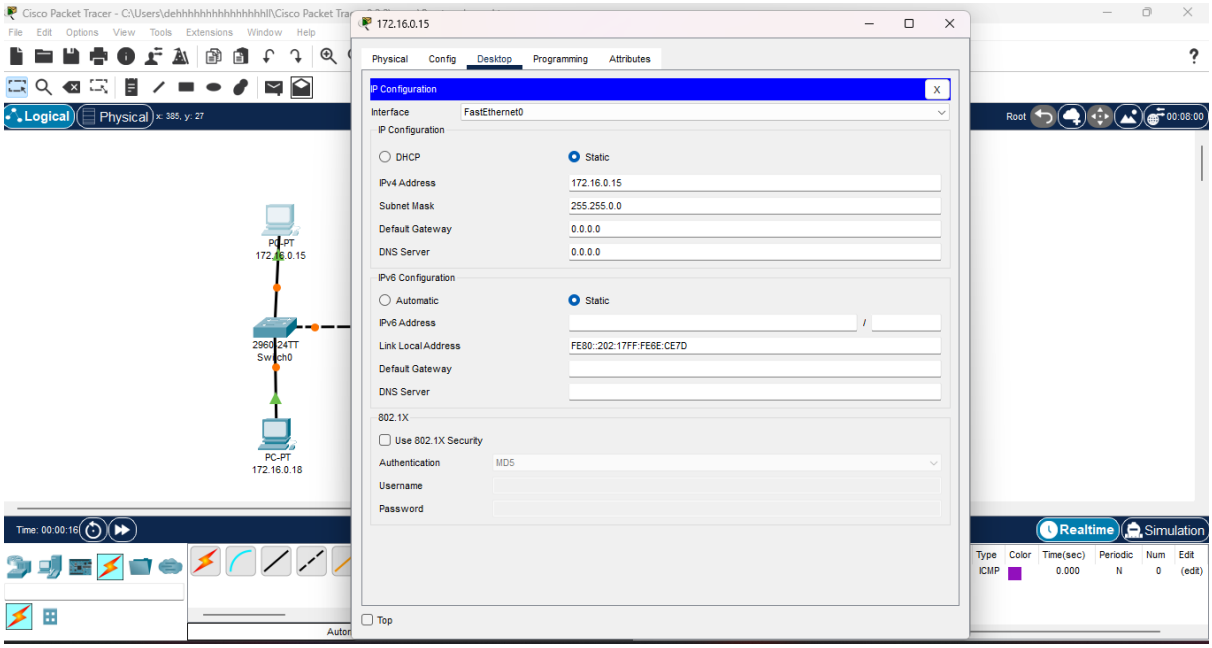
1. Open Packet Tracer
2. Add **PCs**



3. Use **coaxial cable / shared medium**



4. Assign IP addresses manually



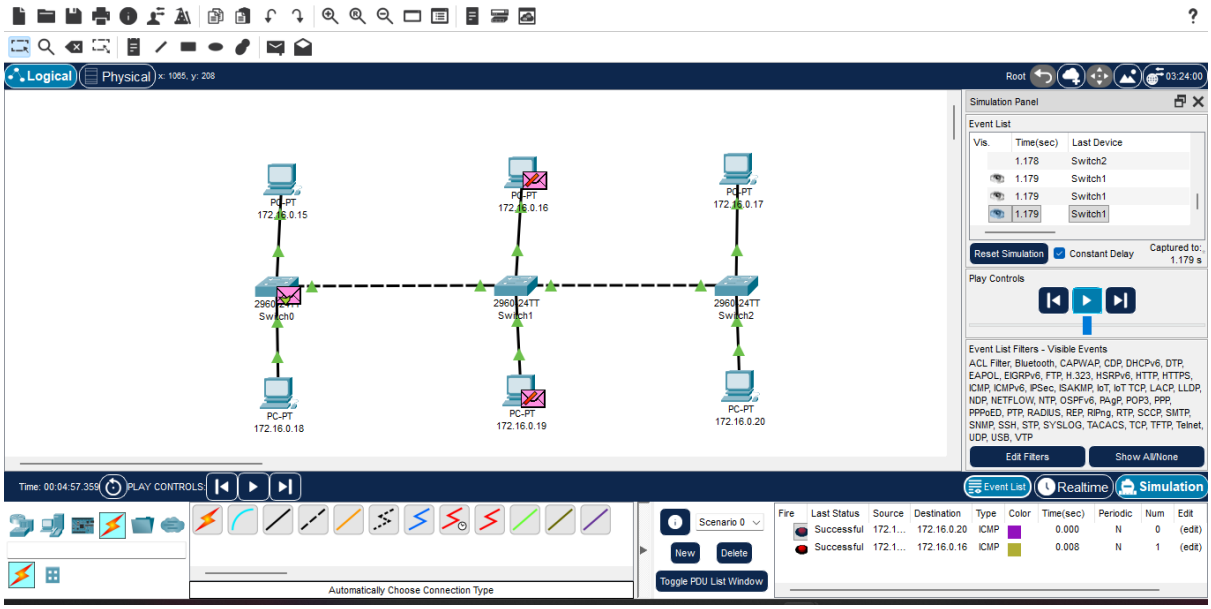
5. Use **ping command** to test communication

172.16.0.15

Physical | Config | Desktop | Programming | Attributes

Command Prompt

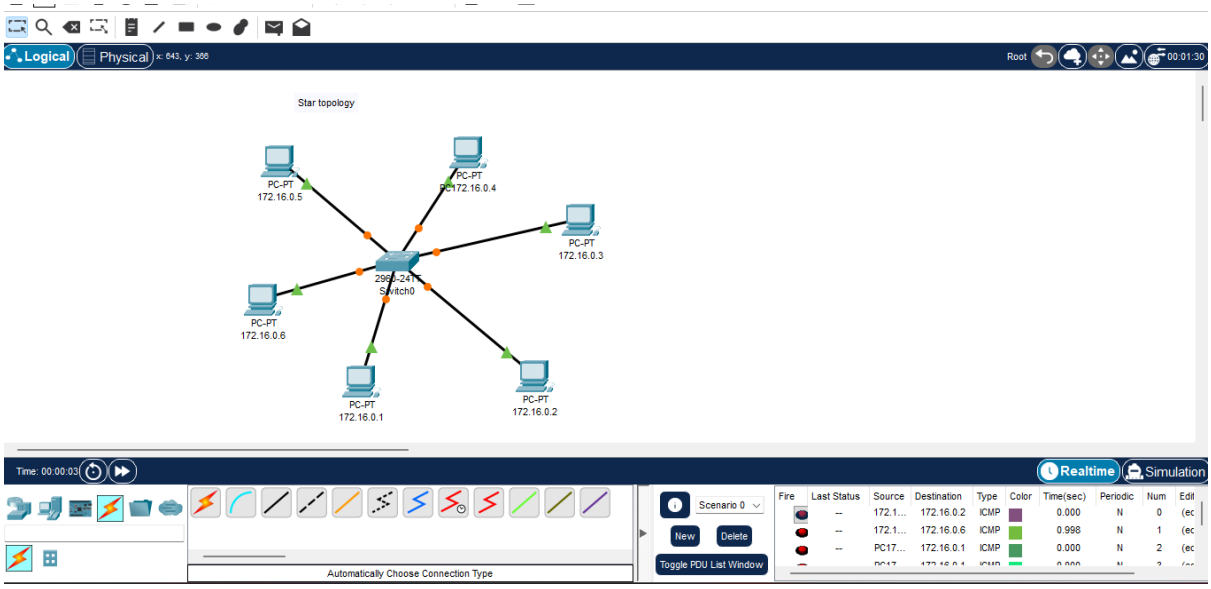
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping172.16.0.18
Invalid Command.

C:\>ping 172.16.0.18

Pinging 172.16.0.18 with 32 bytes of data:

Reply from 172.16.0.18: bytes=32 time=3ms TTL=128
Reply from 172.16.0.18: bytes=32 time<1ms TTL=128
Reply from 172.16.0.18: bytes=32 time<1ms TTL=128
Reply from 172.16.0.18: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>
```



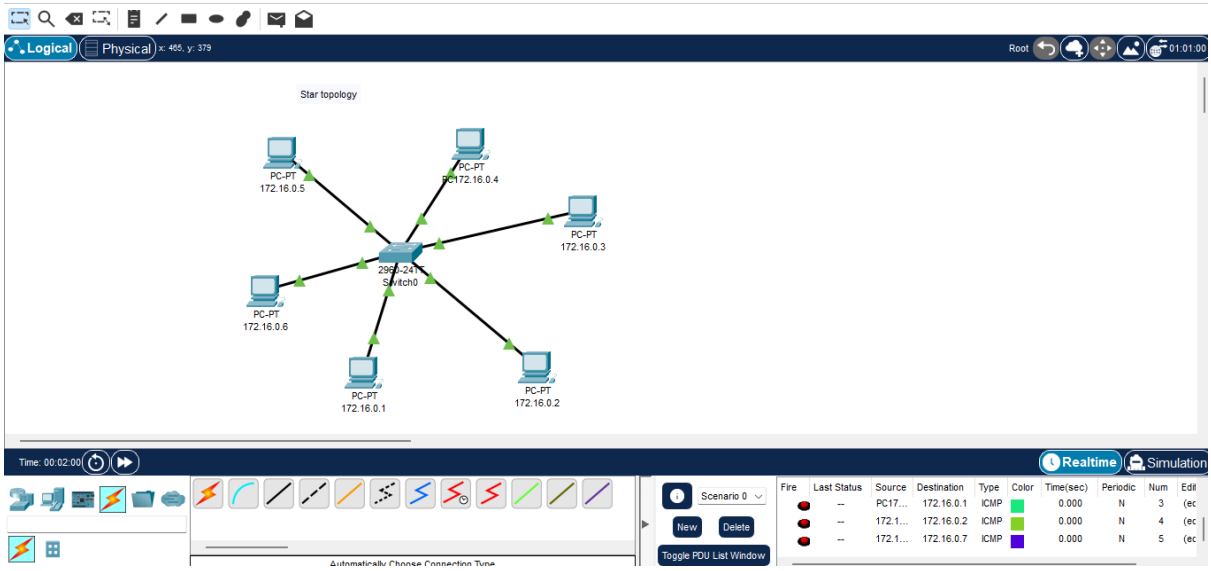
2. **Star topology:-**

Steps:

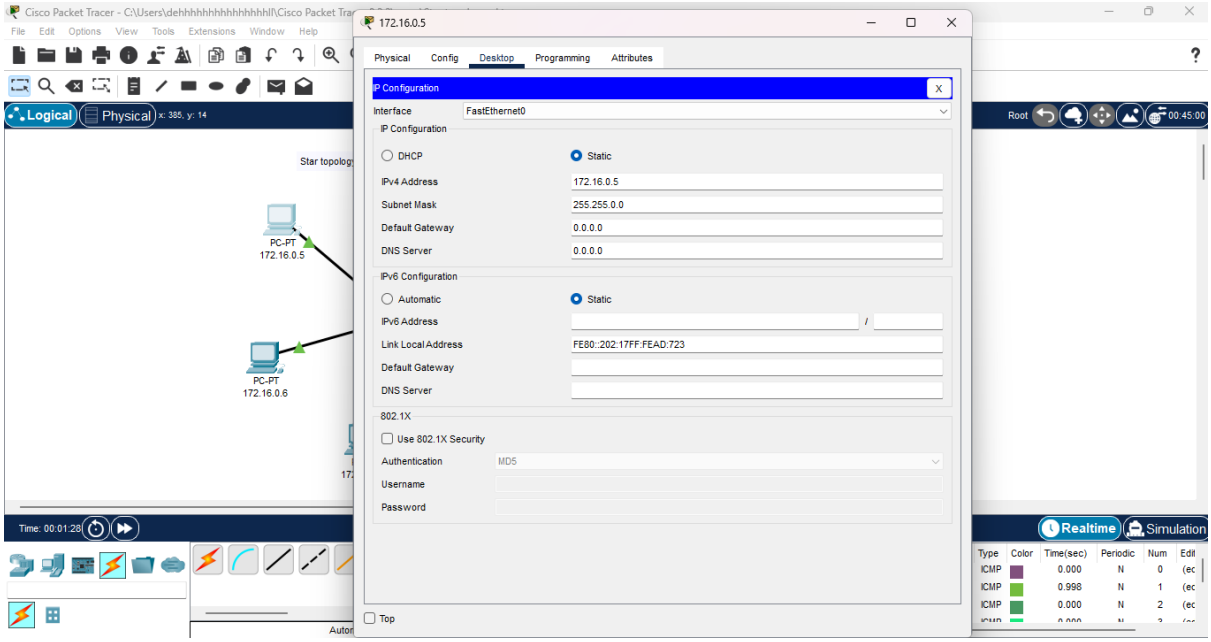
1. Open Cisco Packet Tracer
2. Add **1 Switch** to the workspace



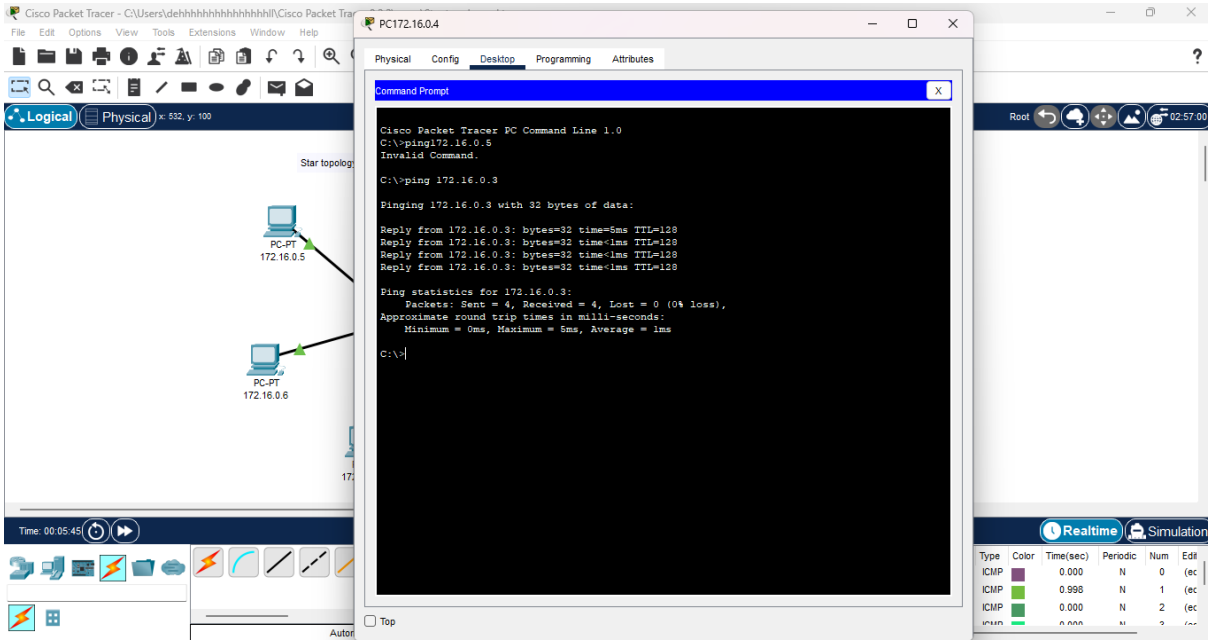
3. Connect each PC to the switch using the cable

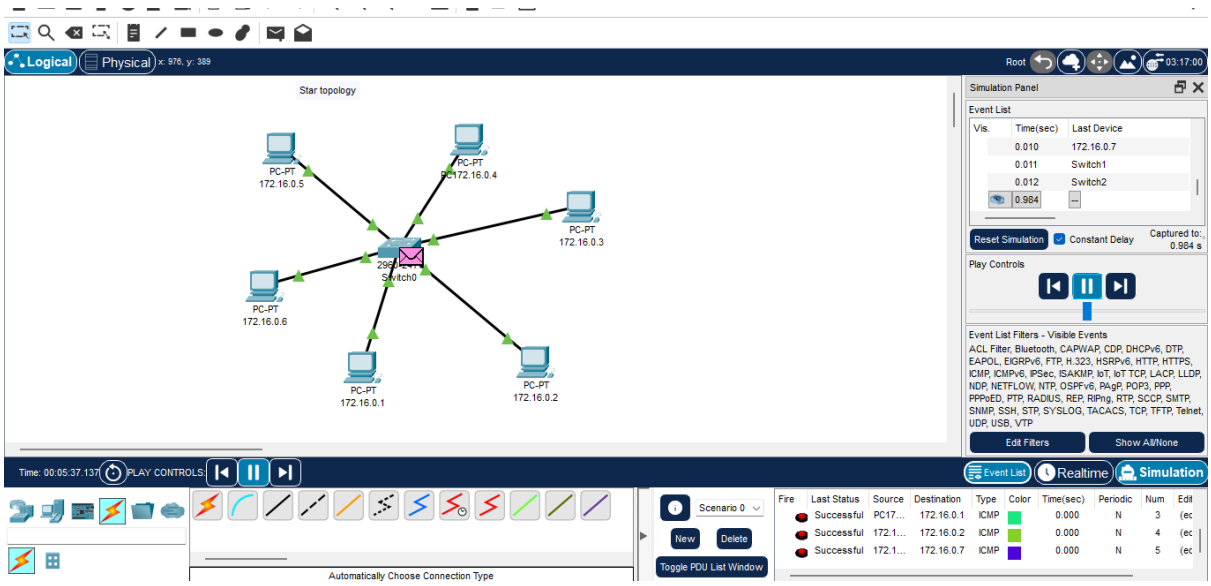


4. Assign **IP addresses** to all PCs



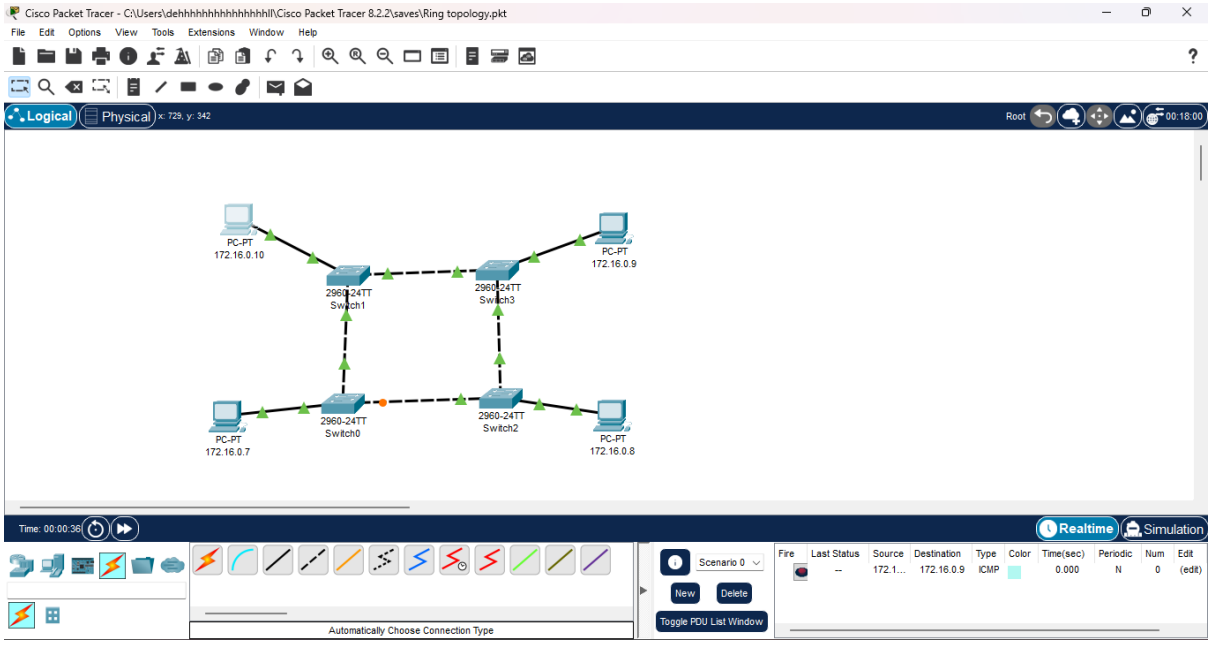
5. Use **ping command** to test connectivity between PCs



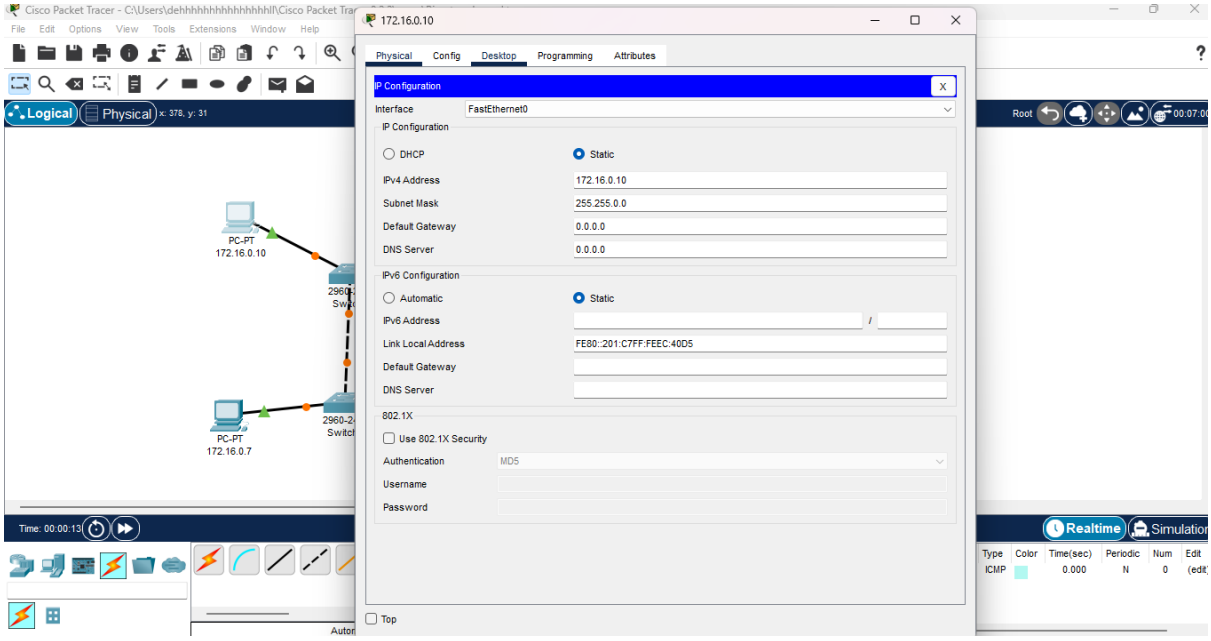


3. **Ring Topology:**

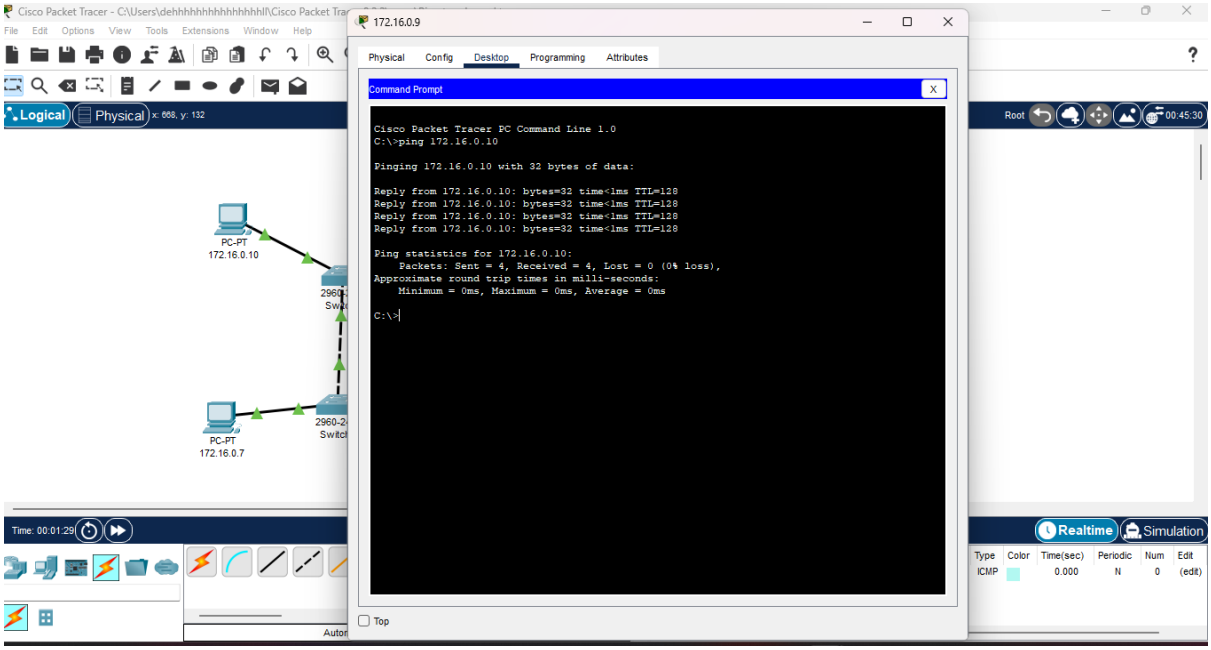
1. Add multiple PCs
2. Connect each PC in circular form



3. Assign IP addresses

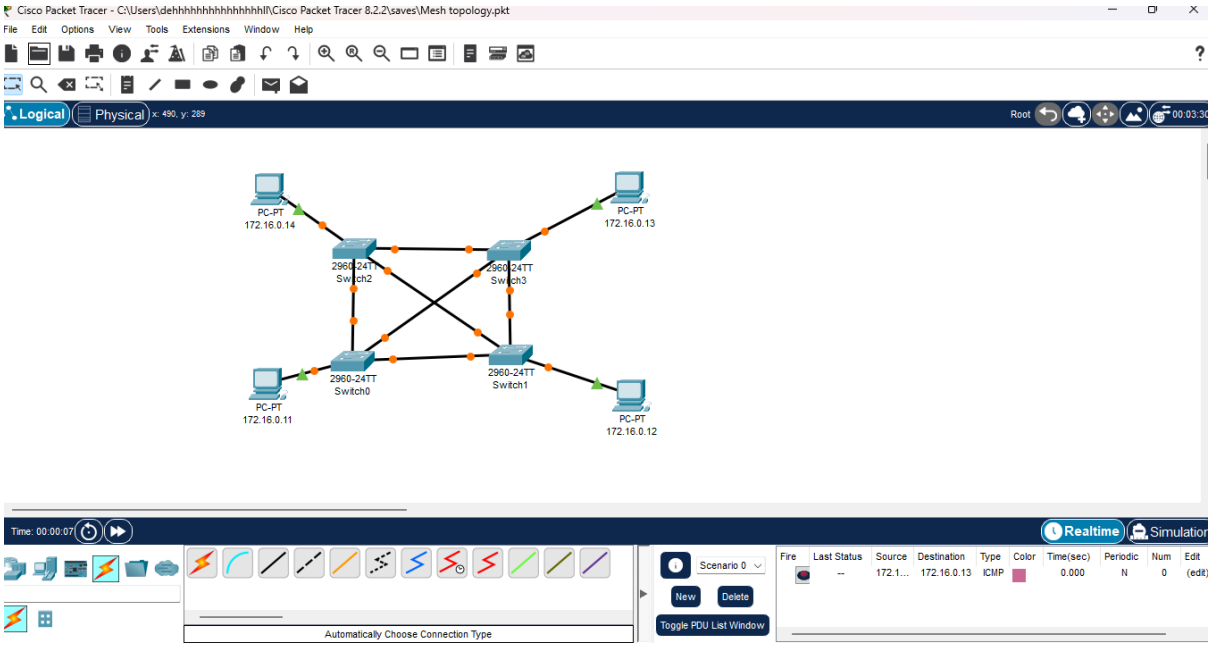


4. Test communication

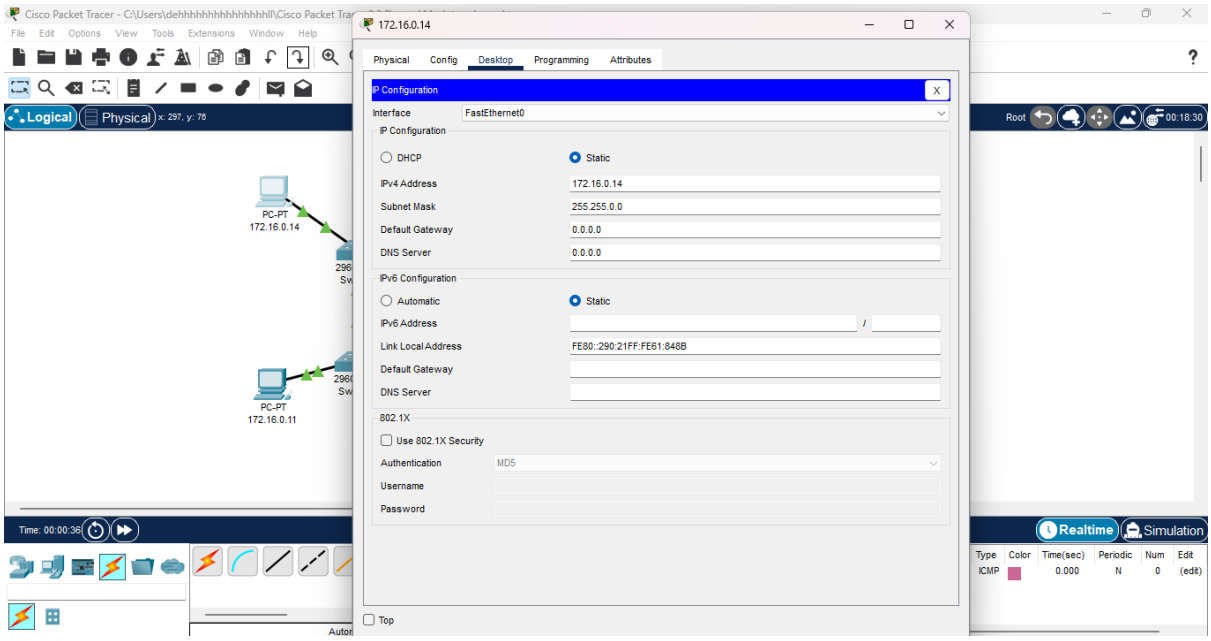


4.Mesh Topology:

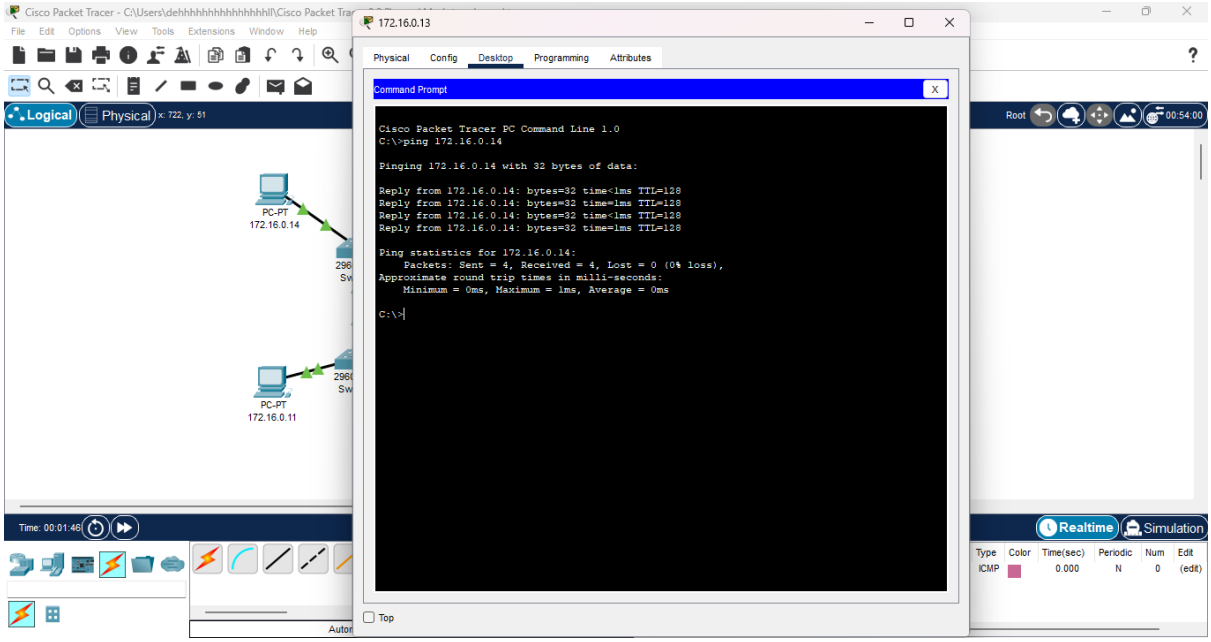
1. Add multiple PCs
2. Connect each device to every other device

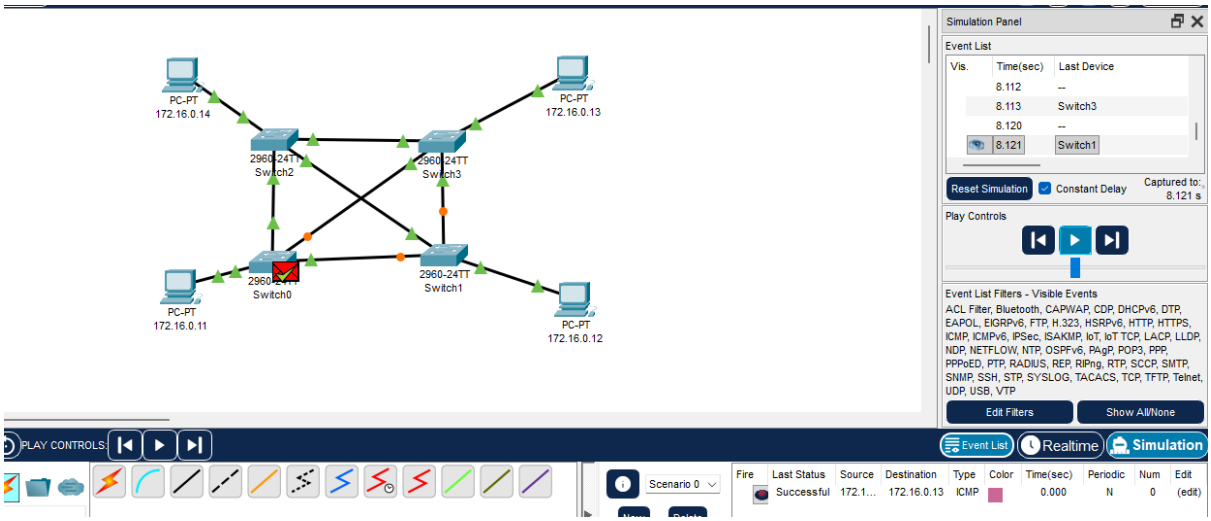


3. Assign IP addresses



4. Test connectivity


172.16.0.13
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.0.14
Pinging 172.16.0.14 with 32 bytes of data:
Reply from 172.16.0.14: bytes=32 time<1ms TTL=128
Reply from 172.16.0.14: bytes=32 time=1ms TTL=128
Reply from 172.16.0.14: bytes=32 time<1ms TTL=128
Reply from 172.16.0.14: bytes=32 time=1ms TTL=128
Ping statistics for 172.16.0.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
PC-PT
172.16.0.14
PC-PT
172.16.0.11
Logical
Physical
Realtime
Simulation


PC-PT
172.16.0.14
PC-PT
172.16.0.13
2960-24TT
Switch2
2960-24TT
Switch3
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
172.16.0.11
PC-PT
172.16.0.12
Simulation Panel
Event List
Vis.
Time(sec)
Last Device
8.112
--
8.113
Switch3
8.120
--
8.121
Switch1
Reset Simulation
Constant Delay
Captured to: 8.121 s
Play Controls
Event List Filters - Visible Events
ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
PLAY CONTROLS:
Event List
Realtime
Simulation
Scenario 0
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Successful
172.1...
172.16.0.13
ICMP
0.000
N
0
(edit)